

Algebra 1

Unit 1

Lesson 6 Homework

Name _____

Date _____

Period _____

1. Explain mathematically why the expressions $\left(a^{\frac{1}{c}}\right)^b$ is equivalent to $(a^b)^{\frac{1}{c}}$ using the expressions $\left(-8^{\frac{1}{3}}\right)^2$ and $(-8^2)^{\frac{1}{3}}$.

2. Identify all of the expressions equivalent to $\left(27^{\frac{2}{3}}\right)^2$.

- $27^{\frac{3}{4}}$
- $\sqrt[4]{27^3}$
- $(27^2)^{\frac{2}{3}}$
- $\sqrt[3]{27^4}$
- $(\sqrt[3]{27})^4$

For questions 3-8, complete each row with equivalent expressions.

	$a^{\frac{b}{c}}$	$\left(a^{\frac{1}{c}}\right)^b$	$(a^b)^{\frac{1}{c}}$	$(\sqrt[c]{a})^b$	$\sqrt[c]{a^b}$
3.		$\left(25^{\frac{1}{2}}\right)^5$			
4.					$\sqrt[3]{(-27)^2}$
5.	$70^{\frac{4}{5}}$				
6.			$(8^7)^{\frac{1}{3}}$		
7.				$(\sqrt{121})^3$	

For questions 8-10, rewrite each expression using a rational exponent in the form $a^{\frac{b}{c}}$.

8. $\sqrt[3]{(-48)^8}$

9. $(\sqrt{18})^3$

10. $\frac{1}{\sqrt[4]{10^5}}$